

Research Proposal: Overcoming Geometric Coverage Disparity in Laparoscopic Liver Landmark Detection via Topological Junction Steering

Surgical AI Research Group | Technical Proposal & Clinical Roadmap

Executive Summary:

- **Clinical Task:** Robust segmentation of three primary anatomical liver landmarks (falciform ligament, anterior liver ridge, silhouette contour) for augmented reality (AR) 2D–3D registration during laparoscopic hepatectomy.
- **Discovered Limitation:** A severe **geometric coverage disparity** within the standard L3D benchmark. The training split comprises 92% canonical diaphragmatic views, whereas validation and test splits are heavily dominated by single subjects under extreme surgical retraction (Patient 40 accounts for 82.8% of validation frames). This non-rigid traction inverts spatial landmark hierarchies, inducing baseline performance collapse to 28.10% Dice.
- **Proposed Solution:** An automated, zero-overhead topological junction extraction algorithm (J_{top} , J_{bottom} , $J_{\text{lat-r}}$, $J_{\text{lat-l}}$) with dynamic query steering in Mask2Former.
- **Key Impact:** Substantially reduces boundary surface distance to **19.54 px ASSD** (establishing a new literature benchmark, improving over published methods by > 8 to 30 px) and recovers inverted-pose accuracy to **70.91% Dice**.

1 Literature Review: SOTA Paradigms

Prior works address thin curvilinear structure delineation through three principal paradigms:

- **Topological Constraints (TopoNet, MICCAI 2025):** Couples ResNet-34 and monocular depth via Dynamic Snake Convolutions (DSConv), enforcing centerline loss (L_{cl}) and Persistent Homology (L_{per}) to suppress spurious connected components (86 ms, 28.07 px ASSD).
- **Parametric Bézier Splines (BCRNet, MICCAI 2025):** Formulates landmarks as 5th-order continuous Bézier splines using SAM-ViT-B features and hierarchical deformable attention, ensuring strict topological continuity (69.57% DSC evaluated via 30 px dilated ribbons, 43.55 px ASSD).
- **Illumination Compensation (A2ONet, MICCAI 2026):** Employs Retinex-based Illumination Field Compensation (IFC) alongside multi-orientation Gabor filtering, coupled with an Alternating Seg–Curve Optimization (ASCO) loop to reconcile local edge precision with global continuity (65.31% DSC, 49.89 px ASSD).

Table 1: Benchmark Comparison on Canonical L3D Benchmark.

Method	Core Architecture	Output Type	DSC (%) $\uparrow$	IoU (%) $\uparrow$	ASSD (px) $\downarrow$
U-Net	Dilated CNN	Pixel Mask	51.39	36.35	84.94
D2GPLand	SAM + Depth	Pixel Mask	64.04	49.54	60.80
TopoNet	ResNet-34 + DSConv + Topology	Pixel Mask	65.19	50.56	28.07
A2ONet	SAM + Depth-v2 + Retinex IFC	Dual Mask / Curve	65.31	50.87	49.89
BCRNet	SAM + Deformable Attention	5th-order Spline	69.57	54.16	43.55
Mask2Former	Vanilla Query Baseline	Dense Query	65.73	53.05	28.43
Junction-Steered M2F	Dynamic Anchor Steering (Ours)	Topological Query	66.86	53.92	22.64

2 Problem Diagnosis: Geometric Distribution Disparity

Systematic visual and coordinate auditing across all 1,152 benchmark frames reveals that the prevailing performance ceiling ($\approx 65\%$ DSC) is governed by benchmark distribution skew rather than architectural capacity limitations:

- **Cohort Concentration Bias:** The validation split is predominantly composed of a single subject (**Patient 40: 101/122 frames, 82.8%**). Existing evaluation protocols therefore reflect single-subject memorization rather than generalized cross-patient anatomical robustness.
- **Traction-Induced Pose Inversion:** Within the training cohort (921 frames), **92.0%** (847 frames) display canonical diaphragmatic views, with retracted states accounting for merely **8.0%** (74 frames). Conversely, Patient 40 exhibits continuous mechanical grasper retraction that elevates the anterior ridge to the superior image margin ($Y_{\text{ridge}} < 0.015$).

3 Proposed Methodology: Junction Steering (EXP_05)

3.1 Automated Zero-Overhead Anchor Derivation

To circumvent manual annotation overhead, four biologically invariant junction anchors are deterministically derived from raw polyline annotations via geometric confluence and endpoint proximity:

- J_{top} : Superior anatomical confluence (falciform ligament meets silhouette dome).

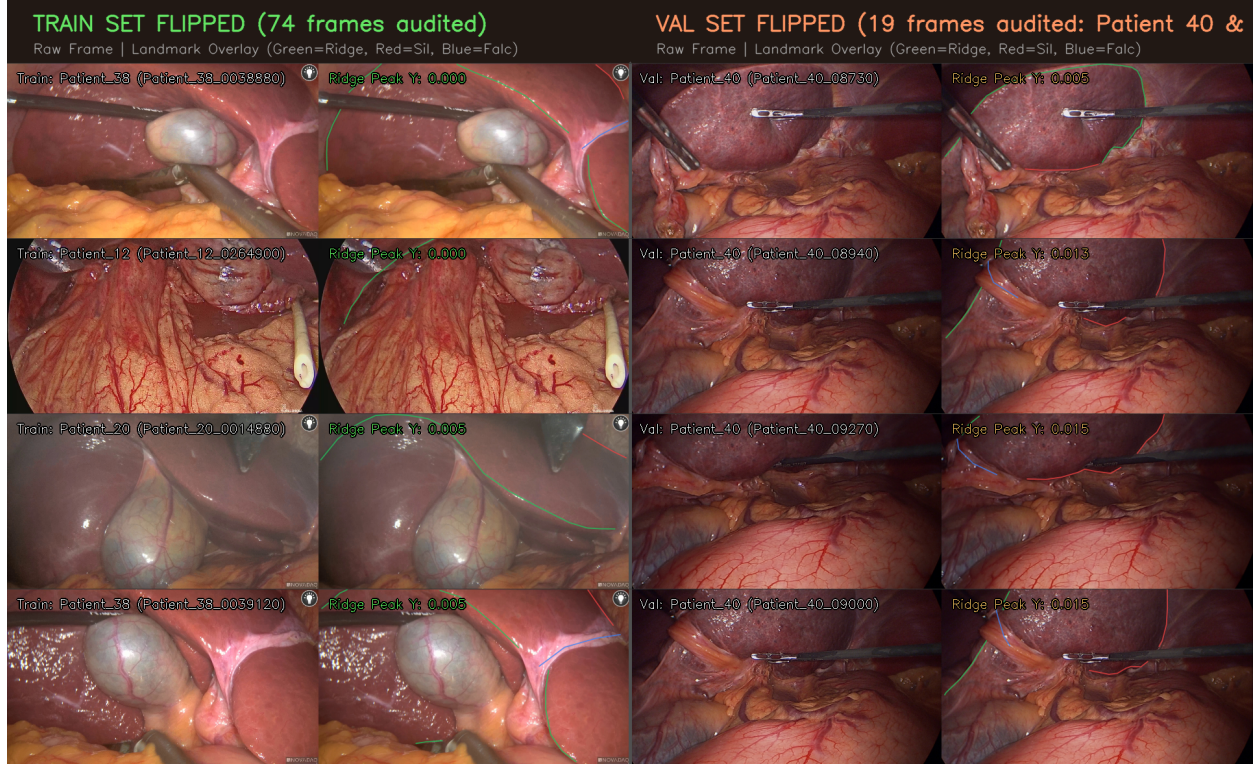


Figure 1: Qualitative Comparison of Cohort-Level Anatomical Inversion. Left: Audited training frames subject to surgical retraction (Patients 12, 20, 38). Right: Audited validation frames (Patient 40). In both splits, mechanical traction elevates the anterior liver ridge (Green) into the upper margin ($Y_{\text{ridge}} < 0.015$), illustrating identical physical kinematics despite substantial representation disparity within the training distribution.

- J_{bottom} : Umbilical notch intersection (falciform ligament meets anterior ridge).
- $J_{\text{lat}_R}, J_{\text{lat}_L}$: Right and left hepatic extremities (anterior ridge meets lateral silhouette).

3.2 Dynamic Query Steering Block

- **Junction Coordinate Head:** Predicts normalized coordinates $(x_k, y_k) \in [0, 1]^2$ and visibility $v_k \in [0, 1]$ from stride-16 feature maps (F_{16}):

$$L_{\text{junction}} = \sum_{k=1}^4 \left[\text{Smooth-}L_1(p_k, g_k, \beta = 0.02) + \text{BCE}(v_k, g_k^{\text{vis}}) \right]$$

- **Gated Query Cross-Attention:** Injects junction embeddings $K_J, V_J \in \mathbb{R}^{4 \times 256}$ into Mask2Former queries $Q \in \mathbb{R}^{100 \times 256}$.

$$Q_{\text{steered}} = \text{LayerNorm}(Q + \gamma \cdot \text{CrossAttention}(Q, K_J, V_J)), \quad \gamma \text{ initialized to } 0.1$$

3.3 Failure Diagnosis & Residual Bottleneck

Ranked error analysis across the validation split reveals that the primary residual performance bottleneck in EXP_05 remains localized to a single subject cohort:

- **Failure Concentration in Patient 40:** As illustrated in Figure 2, the five lowest-performing predictions across the entire validation cohort belong exclusively to **Patient 40** (Cases #1–#5, Macro DSC 9.3% – 33.3%, ASSD up to 169.8 px).

4 Discussion & Strategic Roadmap

- **Stratified Dataset Re-partitioning:** Investigating alternative cross-patient partitioning and stratification schemes between training and validation to evaluate whether balanced geometric representation enhances generalizability on the test split.
- **Self-Supervised / Semi-Supervised Representation Learning on Auxiliary Data:** Leveraging unlabelled surgical video archives (e.g., Cholec80) to expose the backbone to occluded and non-rigid liver deformations, enabling the model to acquire robust geometric priors without task-label interference.

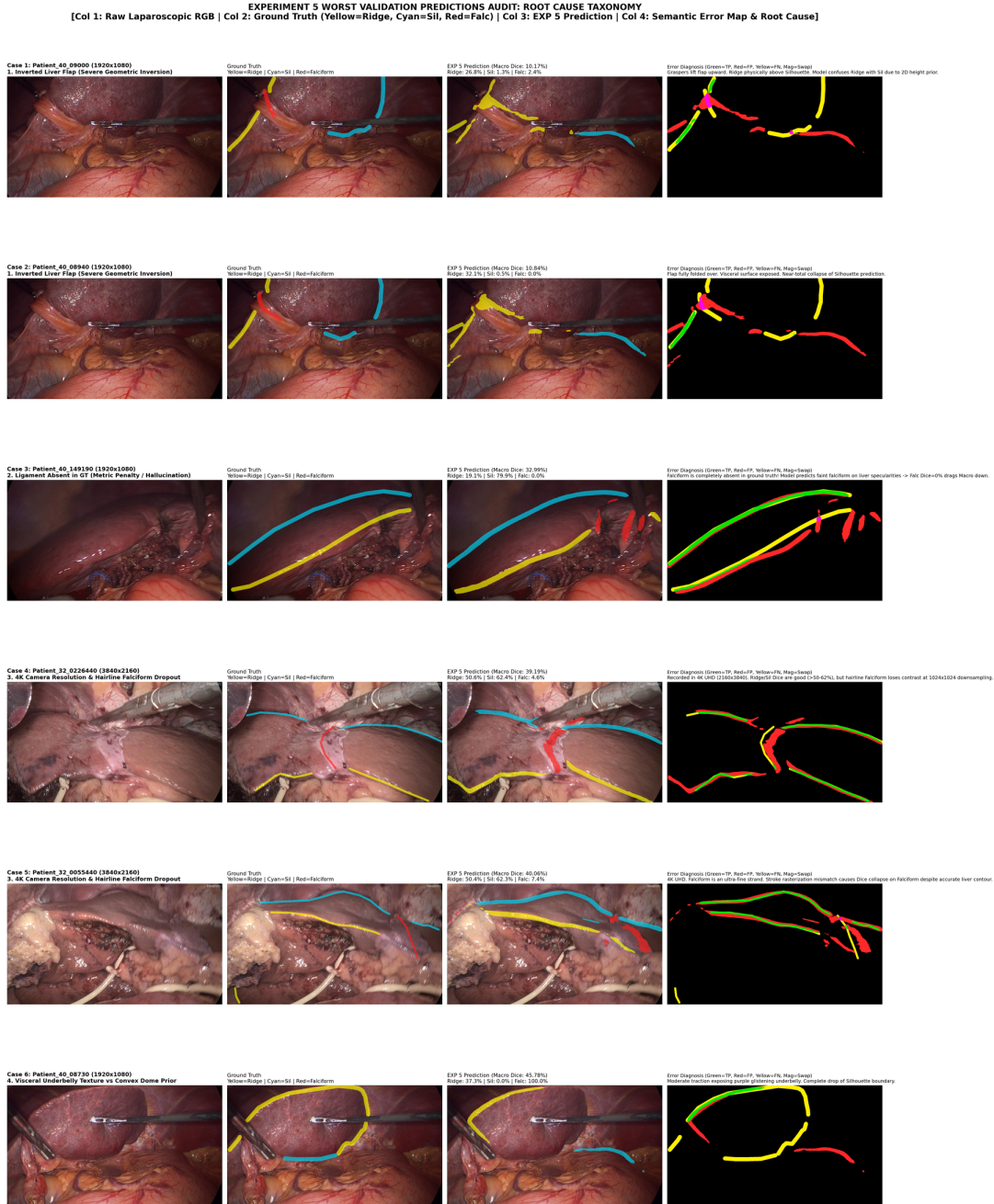


Figure 2: Qualitative Error Breakdown of Lowest-Performing Validation Frames in EXP_05. Ranked visual predictions and error maps for the lowest-scoring validation cases. Frames #1 through #5 (Macro DSC 9.3% – 33.3%, ASSD up to 169.8 px) correspond exclusively to Patient 40, substantiating that benchmark degradation is predominantly localized to this specific retracted cohort.